

Solution Requirements

Date	08 October 2026
Team ID	SWTID-2026-2386
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Not required. EduGenie is an open learning tool with no accounts; no personal data is collected	Low
2	Authorization levels	Single role (student). The Gemini API key is held only by the server and is never sent to the browser	High
3	External interfaces	Google Gemini API (text) through the <code>google-genai</code> SDK	High
4	Transactions processing	Each request is validated, turned into a prompt, sent to Gemini and the result returned. Nothing is stored	High
5	Reporting	Formatted answers, explanations, summaries and learning paths; a three-question quiz with instant right/wrong feedback and the correct answer shown	High
6	Business rules	Input must not be empty and is limited to 8000 characters; a quiz has exactly 3 multiple-choice questions with 4 options and one correct answer; answers stay educational	High
7	Compliance to laws or regulations	API key kept in <code>.env</code> and never committed; no personal data is stored; text typed by the student is sent to the Google Gemini API, so students should not enter private information	Medium
8	Other	Automatic model fallback if the chosen Gemini model is unavailable; friendly error messages with no keys or stack traces	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	EduGenie's own processing is small; Gemini latency dominates	Avg app response < 2 s and max < 5 s for requests that do not call Gemini (measured with <code>tests/perf_load.py</code>)
2	Scalability	Stateless request handling, so more Uvicorn workers can be added; no database to share	25 concurrent users with < 1 % errors on non-Gemini requests
3	Security & Data Privacy	API key in <code>.env</code> , never in code or browser; AI text shown with <code>textContent</code> so it cannot inject HTML; no stack traces shown	No key in the repository or in any response; all errors return a friendly message
4	Reliability & Availability	Falls back through several Gemini models; clear message if Gemini is unreachable	A retired or busy model does not stop the app while another listed model works
5	Usability & Accessibility	One simple page; labelled fields; responsive layout	Usable at 360 px width; keyboard navigable
6	Other	Maintainability and portability: one module per feature, <code>.env</code> configuration, Dockerfile	Runs with <code>pip install -r requirements.txt</code> and <code>uvicorn main:app</code>